10 Testing smart contracts

This chapter covers

- Seeing the importance of testing smart contracts
- Writing test scripts in JavaScript
- Using Truffle frameworks to support smart contract testing
- Interpreting outputs from running test scripts
- Developing test scripts for the counter, ballot, and blind auction smart contracts

This chapter introduces you to a systematic approach for writing test scripts for the smart contracts of your Dapps. Testing is an essential step in any system development process: hardware or software. It is all the more critical for decentralized blockchain applications with unknown peer participants. In chapters 2-9, you tested the integrated Dapp consisting of the smart contract (Dapp-contract) and the app (Dapp-app) by interacting with the web UI. Although this approach is acceptable for testing the functionality of Dapp, systematic testing of a smart contract is required to ensure the robustness of the core logic of a decentralized application. This testing involves exercising every function and every modifier of a smart contract. For this exhaustive and extensive testing, covering all the possible execution paths, inputting the test commands and parameters